

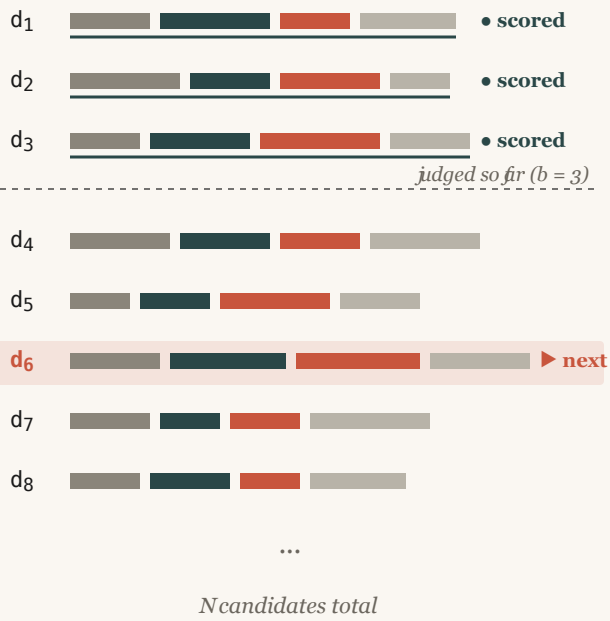
ORE — Online Relevance Estimation

A linear bandit learns which signals matter — per query, online

CANDIDATE POOL

each doc carries a feature vector

■ BM25 ■ dense ■ affinity ■ neighbor



only a few will be sent to the expensive ranker

top-b

LINEAR BANDIT
 $score(d) = \alpha^T \varphi(d)$

PARAMETERS α

α_{BM25}		0.18
α_{dense}		0.42
α_{aff}		0.31
α_{nbr}		0.09

to rerank

EXPENSIVE RANKER
cross-encoder / RankLLaMA / etc.
ground-truth signal